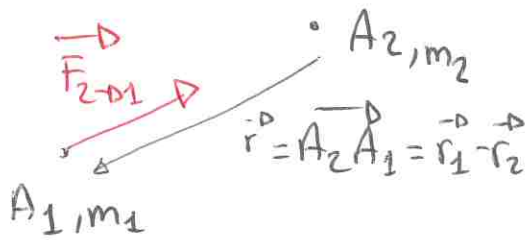


I] Force de gravitation



$$\vec{e}_r = \frac{\vec{r}}{r}$$

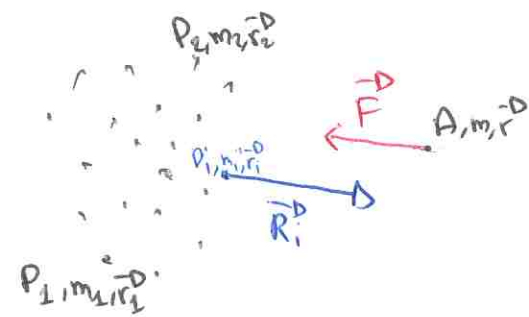
• Force gravitation: $\vec{F}_{2 \rightarrow 1}^{\text{grav}} = G \frac{m_1 m_2}{r^2} \overrightarrow{A_1 A_2} = -G \frac{m_1 m_2}{r^2} \vec{e}_r$

• $G = \text{cst} = 6,67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$

• En EM: $\vec{F}_{2 \rightarrow 1}^{\text{elec}} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \vec{e}_r$

• attractive / répulsive

II] Champ et Potentiel de gravitation



• $\vec{F} = \sum_i \vec{F}_{P_i \rightarrow A} = -G m \sum_i \frac{m_i}{P_i A^2} \times \left(\frac{\overrightarrow{P_i A}}{P_i A} \right)$

$$= -G m \sum_i \frac{m_i}{R_i^2} \left(\frac{\vec{R}_i}{R_i} \right)$$

• $\vec{R}_i = \vec{P}_i A$

$$\vec{g} = \frac{\vec{F}}{m} = -G \sum_i \frac{m_i}{R_i^2} \left(\frac{\vec{R}_i}{R_i} \right)$$

• Si Distrib. masse. continue:

↳ $\rho = \text{masse volumique} \rightarrow \rho dV \text{ masse}$

$$\Rightarrow \vec{g}^D = -G \int_V \rho \frac{\vec{PA}^D}{PA^3} dV = -G \int_V \frac{\vec{R}^D}{R^3} \rho dV$$

• De même: On définit Potentiel de gravitation ϕ :

$$\left[\phi = -G \sum_i \frac{m_i}{P_i A} \right] \quad \text{ou} \quad \left[\phi = -G \int_V \frac{\rho}{PA} dV \right]$$

III] Propriétés du champ de gravitation

On a:

$$\delta W = \vec{F}^D \cdot d\vec{r} = \vec{F}^D \cdot d\vec{R}_i = -Gm \sum_i \frac{m_i}{R_i^2} \left(\frac{\vec{R}_i^D}{R_i} \right) \cdot d\vec{R}_i$$

$\parallel R_i dR_i$

$$\Rightarrow \delta W = -Gm \sum_i \frac{m_i}{R_i^2} dR_i = d \left(mG \sum_i \frac{m_i}{R_i} \right)$$

$-dE_p$

$$\delta W = -dE_p; \quad \left[E_p = -Gm \sum_i \frac{m_i}{R_i} \right] \quad \text{et} \quad \left[\phi = \frac{E_p}{m} \right]$$

$$\vec{F}^D = m\vec{g}^D$$

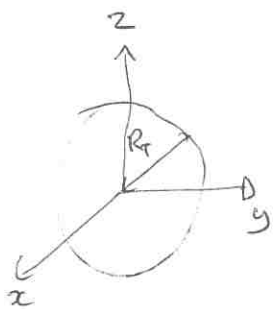
$$E_p = m\phi$$

$$\left. \begin{array}{l} \vec{F}^D \cdot d\vec{r} = -dE_p \\ E_p = m\phi \end{array} \right\} \Rightarrow \left[\vec{g}^D \cdot d\vec{r} = -d\phi \right]$$

$$\Rightarrow \left[\vec{g}^D = -\vec{\text{grad}} \phi \right] \quad \text{et} \quad \left[\oint \vec{g}^D \cdot \vec{n} dS = -4\pi G M_{in} \right]$$

th de Gauss

IV] Calculs de champs et de potentiels



• Extérieur : si $r > R$:

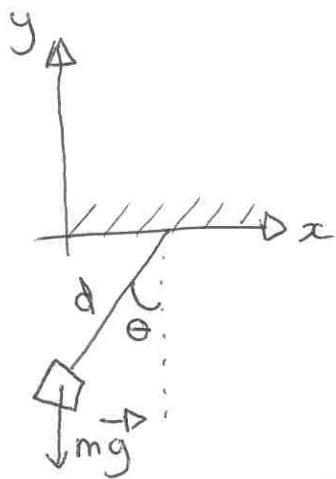
$$F = \oint \vec{g} \cdot d\vec{S} = 4\pi r^2 g(r) = -4\pi G M_{in}$$

$$\Rightarrow \left[\vec{g}(r) = - \frac{G M_T}{r^2} \vec{e}_r \right]$$

$$\vec{g}_p = \frac{G M_T}{R_T^2} \vec{e}_r$$

$$\begin{array}{c} \odot m \\ \downarrow \vec{F} = m \vec{g}_p \end{array}$$

Experience



$$\Rightarrow \ddot{\theta} + \frac{g}{d} \sin \theta = 0$$

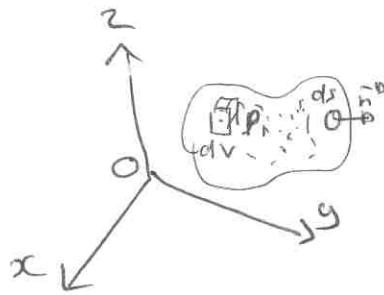
Oscillateur harmonique : $T_0 = 2\pi \sqrt{\frac{d}{g}}$

$$\Rightarrow g = \frac{d \times 4\pi^2}{T_0^2}$$

$$g_{ref} = 9,8087 \text{ m.s}^{-2}$$

• Le flux du champ $\vec{g}^D$... Surface fermée S :

$$F = \oint_S \vec{g}^D \cdot \vec{n}^D ds$$



$$F = -G \oint_S \vec{n}^D \cdot \sum_i m_i \frac{\vec{P}_i \vec{A}^D}{P_i^3} ds = -G \sum_i m_i \oint_S \frac{\vec{n}^D \cdot \vec{P}_i \vec{A}^D}{P_i^3} ds$$

$\approx d\Omega_i$

angle solide

$\Rightarrow \oint d\Omega_i = 4\pi$ pour une
sphère

$$\vec{g}^D = -\vec{\text{grad}} \phi \quad \text{et} \quad \text{div}(\vec{g}^D) = -4\pi G\rho$$

$$\Rightarrow \boxed{\Delta \phi = 4\pi G\rho}$$

$$\rho = 0 \Rightarrow \Delta \phi = 0$$

V] Lois de Kepler

- Planètes décrivent des ellipses
- Balayent des aires égales en temps égaux
- période² $\propto$ demi-grandaxe³

A] Force centrale et 2^{ème} loi de Kepler



$$\vec{M}_O(\vec{F}) = \vec{OM} \wedge \vec{F} = 0$$

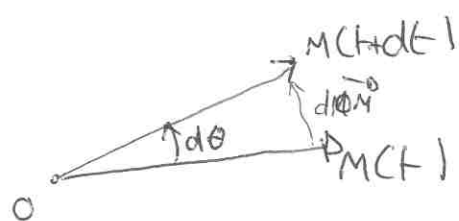
$$\Rightarrow \vec{L}_O = \text{cst} = m \vec{OM} \wedge \vec{v} \stackrel{\text{en polaire}}{=} m r^2 \dot{\theta} \vec{U}_2$$

$$\Rightarrow C = r^2 \dot{\theta} = \text{cst}$$

Mercure : tour complet 88 jours

Neptune : " " 165 ans

• Lien avec vitesse dréolaire :



$$dA = \frac{1}{2} \|\vec{OM} \wedge d\vec{OM}\|$$

$$= \frac{1}{2} \|\vec{OM} \wedge \vec{v}(t) dt\|$$

$$\Rightarrow \frac{dA}{dt} = \frac{1}{2} \frac{\|\vec{L}_O\|}{m} = \frac{|C|}{2}$$

2

B) Forces Centrales Conservatives

$$C = r^2 \dot{\theta} = \text{cst}$$

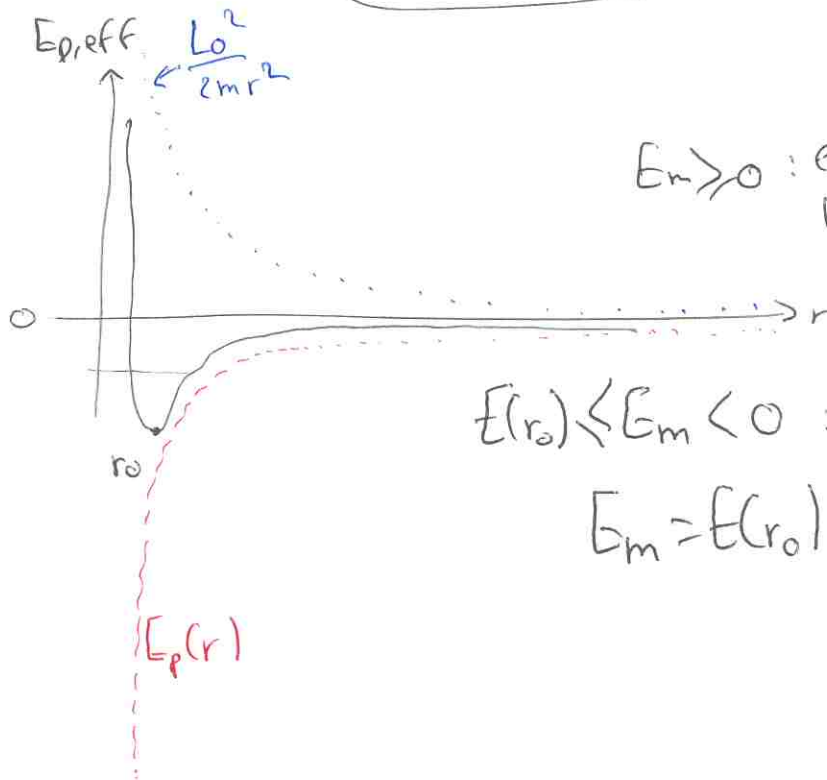
$$\vec{v} = \dot{r} \vec{e}_r + r \dot{\theta} \vec{e}_\theta$$

$$\Rightarrow E_c = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m (r \dot{\theta})^2 = \frac{1}{2} m \dot{r}^2 + \frac{L_0^2}{2 m r^2}$$

$$E_m = E_c + E_p = \left(E_p(r) + \frac{L_0^2}{2 m r^2} \right) + \frac{1}{2} m \dot{r}^2$$

$E_{p, \text{eff}}$ Dépend de r

On a $\vec{F}_g$ de la forme : $\left[\vec{F} = -\frac{k}{r^2} \vec{U}_r \right]$ et $\left[E_p(r) = -\frac{k}{r} \right]$



$E_m > 0$: état de diffusion
↳ tous les "r" accessibles.

$E(r_0) < E_m < 0$: entre $r_{\min}$ et $r_{\max}$

$E_m = E(r_0) \Rightarrow$ cercle

C] Calcul des trajectoires:

$$\text{On a : } m \frac{d\vec{v}}{dt} = - \frac{K}{r^2} \vec{U}_r$$

$$\text{Runge - Lenz : } \boxed{\vec{A} = \vec{v} \wedge \vec{L}_0 - K \vec{U}_r}$$

$$\text{On Rq : } \frac{d\vec{A}}{dt} = \frac{d\vec{v}}{dt} \wedge \vec{L}_0 - K \dot{\theta} \vec{U}_\theta = \frac{-K}{mr^2} \vec{U}_r \wedge mr^2 \dot{\theta} \vec{U}_\theta - K \dot{\theta} \vec{U}_\theta = \vec{0}$$

• On choisit l'axe (Ox) selon $\vec{A}$

$$\Rightarrow \vec{A} \cdot \vec{U}_r = A \cos(\theta) = \frac{m c^2}{r} - K$$

$$\Rightarrow \boxed{r = \frac{m c^2 / K}{1 + \frac{A}{K} \cos \theta}}$$

$$\text{Eq: Conique } r = \frac{p}{1 + e \cos \theta}$$

• ~~Si $e = 0$:~~

• $K > 0$:

$$\Rightarrow E_m = \frac{K}{2p} (e^2 - 1)$$

• si $e > 1$: $E_m > 0$: $e=1$: Parabole
 $e > 1$: hyperbole

• si $0 < e < 1$: $E_m < 0$: Ellipse

• si $e = 0$: $E_m = 0$: Cercle

①

$$L_0^2 = m K a (1 - e^2)$$

$$a = \frac{p}{1 - e^2} ; b = \sqrt{a p}$$

Terre: $e = 0,0167 \sim$ Quasi Circulaire 6

• Si trajectoire fermée:

: Période Rotation = T

Aire Balayée durant T = aire Ellipse

$$\text{Aire Balayée} = \left(A = \frac{c}{2} T \right) \quad \left\| \quad k = GMm \right.$$

$$\text{Aire Ellipse} : \left(\pi ab \right) \quad \left\| \quad \frac{cT}{2} = \pi ab \right.$$

$$\left(\frac{T^2}{a^3} = \frac{4\pi^2}{GM} \right)$$

③

